

## MATRICES AND DETERMINANTS

### MATRIX

A rectangular array of  $mn$  numbers in the form of  $m$  horizontal lines (called rows) and  $n$  vertical lines (called columns), is called a matrix of order  $m$  by  $n$ , written as  $m \times n$  matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

### TYPES OF MATRICES

#### Zero Matrix or Null Matrix

A matrix each of whose elements is zero, is called a zero matrix or a null matrix.

#### Square Matrix

A matrix in which number of rows is equal to the number of columns, say  $n$ , is called a square matrix of order  $n$ .

#### Diagonal Matrix

A square matrix  $A = [a_{ij}]_{n \times n}$  is called a diagonal matrix if all the elements except those in the leading diagonal are zero, i.e.,  $a_{ij} = 0$  for  $i \neq j$ . In other words

$$A = \text{diag.}[a_{11} \quad a_{22} \quad a_{33} \quad \cdots \quad a_{nn}]$$

#### Unit Matrix

A square matrix in which every non-diagonal element is zero and every diagonal element is 1, is called a unit matrix or an identity matrix. Thus, a square matrix  $A = [a_{ij}]_{n \times n}$  is a unit matrix if

$$a_{ij} = \begin{cases} 0 & \text{when } i \neq j \\ 1 & \text{when } i = j \end{cases}$$

## ALGEBRA OF MATRICES

### Addition of Matrices

Let A and B be two matrices each of order  $m \times n$ . Then the sum matrix  $A + B$  is defined only if matrices A and B are of same order. The new matrix, say  $C = A + B$  is of order  $m \times n$  and is obtained by adding the corresponding elements of A and B.

### Subtraction of Matrices

Let A and B be two matrices of the same order. Then by  $A - B$ , we mean  $A + (-B)$ . In other words, to find  $A - B$  we subtract each element of B from the corresponding element of A.

### Multiplication of Matrices

Two matrices A and B can be multiplied only if the number of columns in A is same as the number of rows in B

## TRANSPOSE OF A MATRIX

Let A be an  $m \times n$  matrix. Then, the  $n \times m$  matrix obtained by interchanging the rows and columns of A is called the transpose of A, and is denoted by  $A'$  or  $A^1$ . Thus,

- (i) if order of A is  $m \times n$ , then, the order of  $A'$  is  $n \times m$ .
- (ii) (i, j)th element of A = (j, i)th element of  $A'$ .

For example, if  $A = \begin{bmatrix} 2 & -3 & -1 \\ 4 & 2 & 3 \end{bmatrix}$ , then  $A' = \begin{bmatrix} 2 & 4 \\ -3 & 2 \\ -1 & 3 \end{bmatrix}_{3 \times 2}$

## SYMMETRIC MATRIX

A square matrix  $A$  is said to be symmetric if  $A' = A$ . That is, the matrix  $A = [a_{ij}]_{n \times n}$  is said to be symmetric provided  $a_{ij} = a_{ji}$  for all  $i$  and  $j$ .

## SKEW SYMMETRIC MATRIX

A square matrix  $A$  is said to be skew symmetric, if  $A' = -A$ . That is, the matrix  $A = [a_{ij}]_{n \times n}$  is skew-symmetric if  $a_{ij} = -a_{ji}$  for all  $i$  and  $j$ .

## ORTHOGONAL MATRIX

A square matrix of order  $n \times n$  is said to be orthogonal if

$$AA' = I_n = A'A.$$

## MINOR

If  $m - p$  rows and  $n - p$  columns from matrix  $A_{m \times n}$ , are removed, the remaining square submatrix of  $p$  rows and  $p$  columns is left. The determinant of a square submatrix of order  $p \times p$  is called a minor of  $A$  of order  $p$ .

(i) every element of the matrix is the minor of order.

(ii)  $\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}, \begin{vmatrix} 3 & 6 \\ 2 & 0 \end{vmatrix}, \begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix}, \begin{vmatrix} 0 & 4 \\ 1 & 0 \end{vmatrix}$  etc. are minors of order 2.

(iii)  $\begin{vmatrix} 3 & 1 & 0 \\ 1 & 3 & 6 \\ 1 & 2 & 0 \end{vmatrix}, \begin{vmatrix} 2 & 1 & 0 \\ 4 & 3 & 6 \\ 8 & 2 & 0 \end{vmatrix}, \begin{vmatrix} 2 & 3 & 1 \\ 4 & 1 & 3 \\ 8 & 1 & 2 \end{vmatrix}$  etc. are the minors of order 3.

## RANK OF A MATRIX

A positive integer  $r$  is said to be the rank of a non zero matrix  $A$ , if

- (i) there exists atleast one minor in  $A$  of order  $r$  which is non zero,
- (ii) every minor in  $A$  of order greater than  $r$  is zero,  $r$  is written as  $\rho(A) = r$ .

The rank of a zero matrix is defined to be zero.

### Properties of Rank of a Matrix

- (i) Rank of a matrix remains unaltered by elementary transformations.
- (ii) No skew-symmetric matrix can be of rank 1.
- (iii) Rank of matrix  $A$  = Rank of matrix  $A'$ .

## SOLUTION OF A SYSTEM OF LINEAR EQUATIONS BY MATRIX METHOD

Consider a system of linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \vdots & \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n &= b_n \end{aligned}$$

We can express these equations as a single matrix equation

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$A \qquad X \qquad B$

Let  $|A| \neq 0$ , so that  $A^{-1}$  exists uniquely. Pre-multiplying both sides of  $AX = B$  by  $A^{-1}$ , we get

$$A^{-1}(AX) = A^{-1}B \quad \text{or} \quad (A^{-1}A)X = A^{-1}B$$

OR  $IX = A^{-1}B$  OR  $X = A^{-1}B$ .

Hence  $X = A^{-1}B$  is the unique solution of  $AX = B$ ,  $|A| \neq 0$ .

## **DETERMINANTS**

### **MINORS AND COFACTORS**

#### **Minor of an Element of a Determinant**

If we take an element of the determinant and delete the row and the column containing that element, the determinant left is called the minor of the element. It is denoted by  $M_{ij}$

For example, given the  $3 \times 3$  determinant

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

Then the minor of  $a_{11}$  of  $M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$ ;

the minor of  $a_{12}$  is  $M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$  and so on.

#### **Cofactor of an Element of a Determinant**

The cofactor  $C_{ij}$  of an element  $a_{ij}$  (the element in the  $i$ th row and  $j$ th column) is defined as  $C_{ij} = (-1)^{i+j}M_{ij}$ . It is denoted by  $C_{ij}$ . Thus,

$$C_{ij} = \begin{cases} M_{ij}, & \text{when } i + j \text{ is even} \\ -M_{ij}, & \text{when } i + j \text{ is odd} \end{cases}$$

For example, the cofactor of  $a_{12}$  in the  $3 \times 3$  determinant

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\text{is } c_{12} = (-1)^{1+2} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}.$$

## PROPERTIES OF DETERMINANTS

1. The value of the determinant remains unchanged if rows are changed into columns and columns are changed into rows,
2. If two adjacent rows (columns) of a determinant are interchanged, the value of the determinant so obtained is the negative of the value of the original determinant, i.e.
3. If two rows or columns of a determinant are identical then its value is zero, i.e.,
4. If each element of a row or column of a determinant is multiplied by a constant  $k$  then the value of the new determinant is  $k$  times the value of the original determinant, i.e.
5. If any two rows or columns of a determinant are proportional, then its value is zero, i.e.

## SOLUTION OF LINEAR EQUATIONS BY DETERMINANTS

### 1. Cramer's Rule: Solution of system of linear equations in two unknowns

The solution of the system of equations

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

is given by  $x = \frac{D_1}{D}$  and  $y = \frac{D_2}{D}$ , where

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, D_1 = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$$

and  $D_2 = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$ , provided  $D \neq 0$ .

## 2. Cramer's Rule: Solution of system linear equations in three unknowns

The solution of the system of equations

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

is given by

$$x = \frac{D_1}{D}, y = \frac{D_2}{D} \text{ and } z = \frac{D_3}{D}, \text{ where}$$

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}; D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}; D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

provided  $D \neq 0$ .

- (i) If  $D \neq 0$ , the system has only trivial solution ( $x = y = z = 0$ ), and
- (ii) If  $D = 0$ , the system has infinitely many solutions.